The sun as colliding beam element breeder accelerator and cosmic ray atmosphere creator

Richard Talman
Laboratory for Elementary-Particle Physics
Cornell University, Ithaca, NY, USA

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- **A theory of cosmic ray production in the solar system**
- Extra-galactic cosmic rays are needed only for initial start-up
- How can positrons and anti-protons be present even at very high energies ?
- **Initially**, particles and anti-particles are captured by the sun's **magnetic** field, into counter-circulating low energy beams.
- **Later**, at higher energy, the magnet field bending becomes negligible compared to the **gravitational** bending;
- Positrons and anti-protons are produced in collisions beween counter-circulating beam particles.
- Both positive and negative beams, having survived the gradual transition from magnetic to gravitational bending.
- Aside: this requires a serious "**fine-tuning**" of the sun's bending fields.

3 Abstract (cont)

- The only role for extra-galactic nuclear particles is to "start up" the solar process from scratch,
- Eventually the self-sustaining atmosphere of cosmic rays currently detectable within the solar system is restored.
- Perhaps, but not necessarily, this start-up needs to be repeated every 11 years, coinciding with solar magnet and electric field reversals.
- Certainly the cosmic ray production must have been started up at least once in the past.
- The cosmic ray energy inherited from outside the solar system is negligiby small compared to the energy povided by the sun.

4 Abstract (cont)

- High quality cosmic ray data has neen collected over recent decades, at steadily increasing energies, especially by the International Space Station (ISS).
- Longitudinal electric fields, explained by the Parker theory of the solar wind, enable the sun to serve as a "booster" accelerator of cosmic rays, increasing cosmic ray energies by many orders of magnitude.
- High energy particle collision processes maintain the particle abundances at every energy.
- The following slides mainly show figures from the full paper.

5 Solar Accelerator

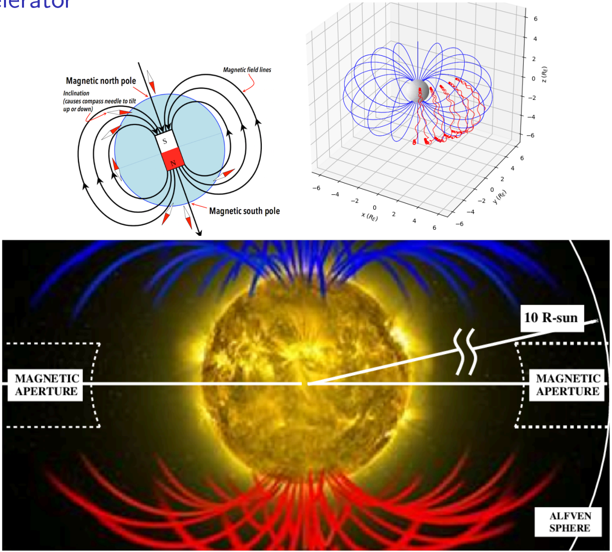


Figure 1: **Top:** Magnetic dipole field pattern views. **Bottom:** The sun's magnetic field lines, with the sun as a particle accelerator.

6 Parker solar wind magnetic and electric field patterns

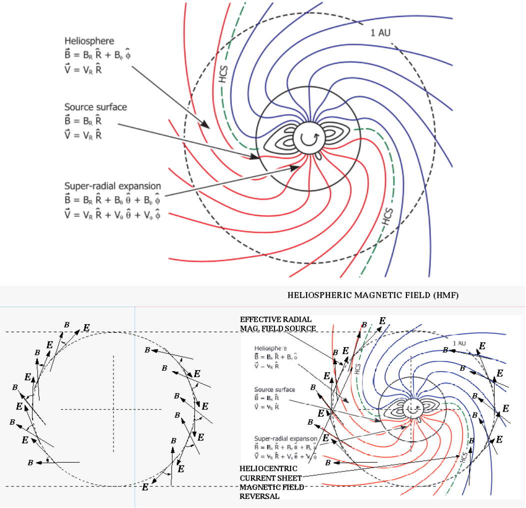


Figure 2: **Top:** Copied from Owens and Forsythe, with **Bottom:** annotated field patterns.

7 Alfven surface location determination

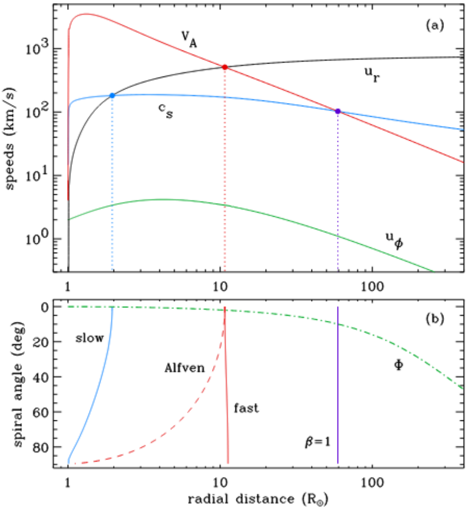


Figure 3: Figure copied from Crammer, calculations showing the (erratic) radius of the Alfven surface at aproximately 10 times the solar radius. β is not the ratio to the speed of light.

8 Experimental Apparatus (for thought experiment)

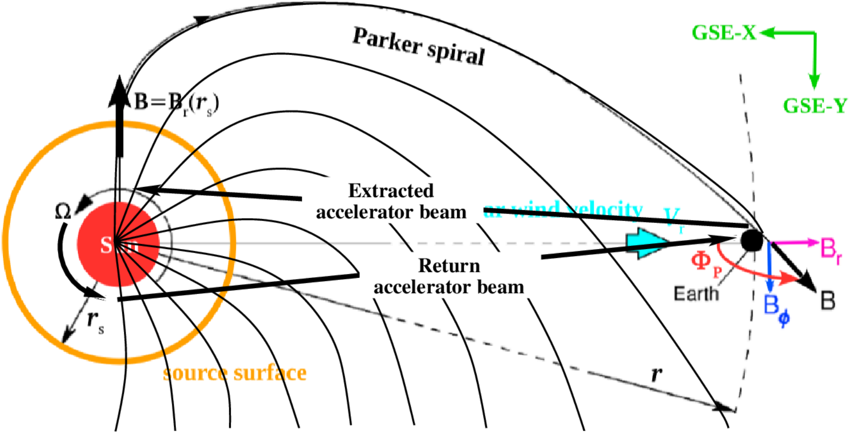


Figure 4: Experimental "Apparatus". The FNAL Tevatron beam is bent up by 5 degrees and pointed toward the sun. The protons are expected back in half an hour. But they don't show up? Precession of the elliptical orbit has not been taken into account.

9 Proton trip from and to a remote source

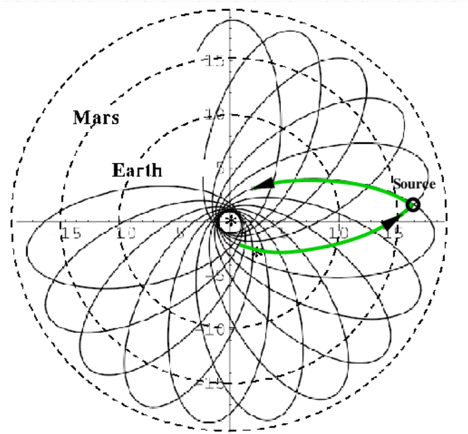


Figure 5: Stable bound orbit for a 6.6 GeV proton traveling **at velocity 0.99 c**, from and to a remote source. Figure copied and annotated from a paper by Munoz and Pavic, doi:10.1088/0143-0807/27/5/001.

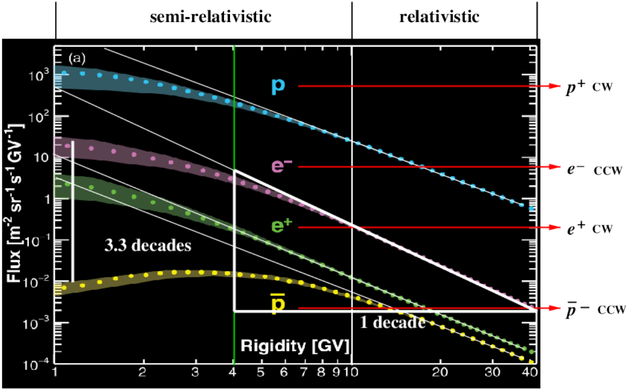


Figure 6: Figure copied from AMS paper showing rigidity (a.k.a. momentum over charge) dependence and relative abundances of protons, electrons, positrons, and anti-protons measured by the AMS specrometer on the ISS. The protons have become fully relativistic only above 10 GeV, while the electrons have are fully relativistic midway through the region labeled "semi-relativistic".

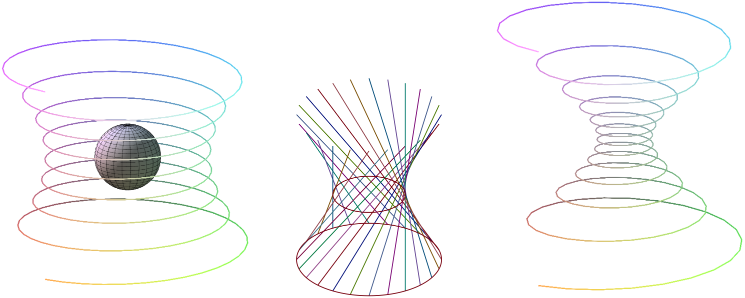


Figure 7: Selected topologies of particles falling into the sun: **Left:** cork-screw shaped orbit, captured briefly, then released; **Center:** grazing the sun; **Right:** falling into the sun.

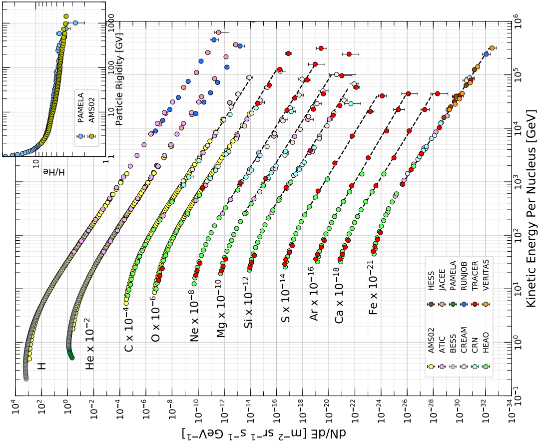


Figure 8: Figure copied from Particle Data Group, Chap-29, Cosmic-Rays. showing energy dependence and relative abundance ratios measured by AMS within the ISS. Some element abundances, such as the “light elements” (Li, Be, B) are not shown; presumably because these nuclei are so rare as cosmic rays.

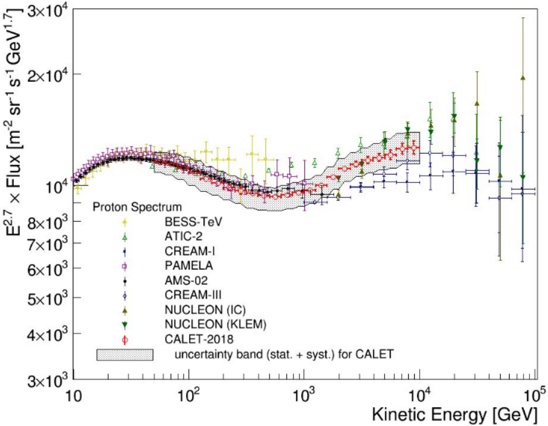


Figure 9: ISS measured flux of cosmic ray protons, plotted as function of proton energy: $\mathrm{dN/dE}$ flux dependence, with best fit exponential factor $\mathrm{KE}^{2.7}$ scaling;

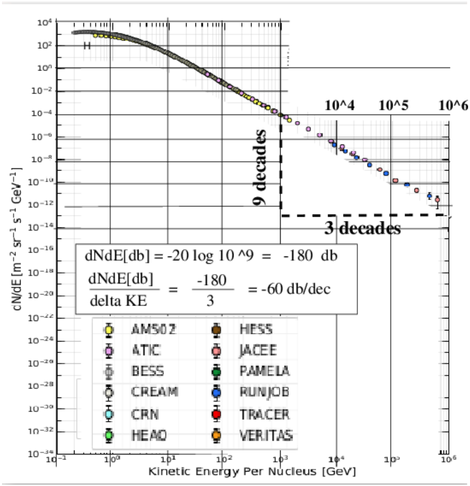


Figure 10: ISS measured flux of cosmic ray protons, plotted as function of proton energy: unscaled data, with best fit exponential decay. This slope of (approximately) 3 decades per decade only matches the fully relativistic data, which is more or less consistent with the $\mathcal{E}^{2.7}$ energy scaling.

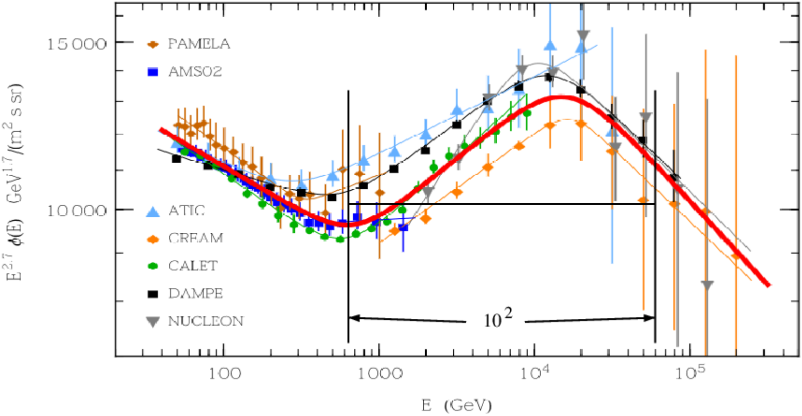


Figure 11: This figure has been copied from Lipari and Vernetto[?]. This data shows that all seven experiments on the ISS were subjected to the same impulse, or that all protons were subjected to the same impulse when near the sun. In the present paper this is interpreted as acceleration of protons from 600 GeV, to 60,000 GeV, while briefly circulating around the sun.

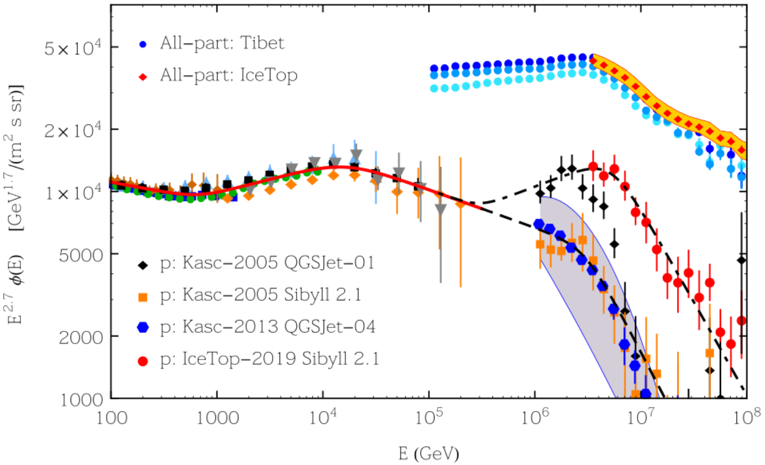


Figure 12: Also copied from Lipari and Vernetto[?], the data in this plot, increase the acceleration range, starting from 100 GeV and running almost to 10^7 GeV.

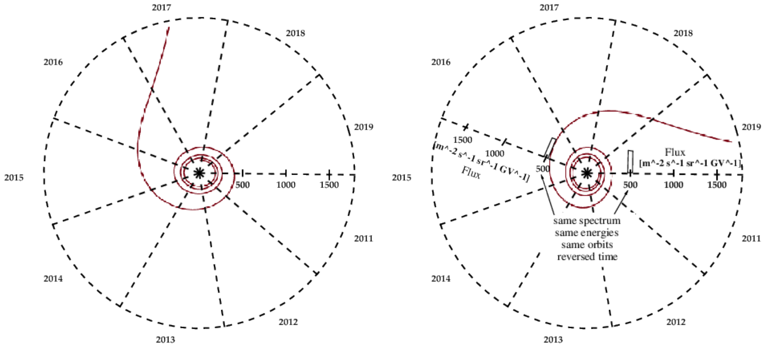


Figure 13: Calculated (pseudo)-orbit for a proton departing from AMS for the sun in mid 2017. The spiral orbits are calculated analytically lituus-modified spirals.

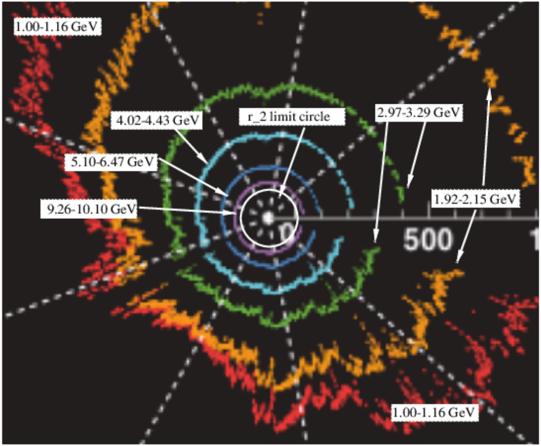


Figure 14: Central portion of a figure from M. Agular, et al. [?]. showing the daily AMS proton fluxes for six typical rigidity bins from 1.0 to 10.10 GV measured from May 20, 2011 to October 29, 2019 which includes a major portion of solar cycle 24 (from December 2008 to December 2019). The AMS data cover the ascending phase, the maximum, and

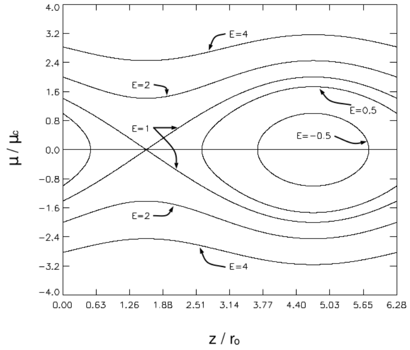


FIG. 1.—Contour plots of the particle energy. Note the presence of a separatrix at $E = 1$ and the trap region for $E < 1$.

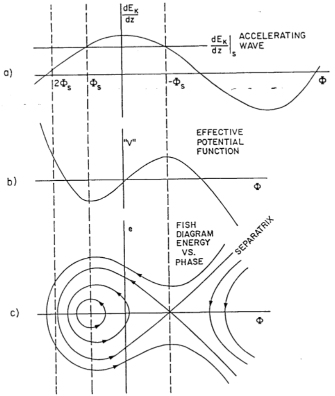
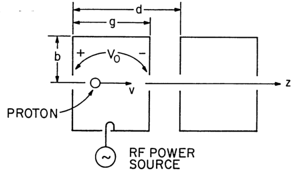


Fig. 6. Phase stability graphs.

Figure 15: **Left:** Copied from Felice and Kulsrud along with its original caption, longitudinal phase space in plasma atmosphere of a star, and **Right:** copied from Loew and Talman, longitudinal phase space in a linear proton "Alvarez" accelerator. These two figures suggests that the physics of the acceleration of cosmic rays near a star resembles the physics of acceleration of electrons in a linear accelerator. In each case there can be charges trapped in moving "stable buckets".



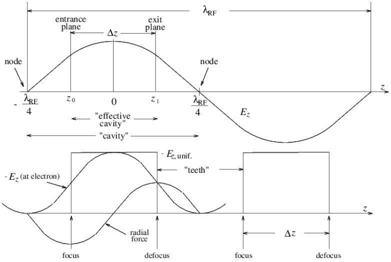


Figure 16: Copied from Loew and Talman, the **left** figure shows one of the many RF-cavity accelerating cells of an Alvarez linac needed to produce a “slow” (meaning slower than the speed of light) electromagnetic traveling wave. The **right** figure resembles Figure 12.1 in the Kudsrud book, which shows plasma fields capable of accelerating cosmic rays in galactic magneto-hydrodynamic Alfven waves. This figure, applicable to cells in an Alvarez linac, models a proton’s instantaneous position and replacement of actual accelerating field by a train of uniform field “teeth” of thickness $\Delta z \simeq \lambda_{\rm RF}/4$.